Software Construction and Development SE-312

LECTURE # 04

Levels Of Process Definition

Course Instructor: Engr. Ruqqiya Asad

Introduction

A **software process** is a structured set of activities that helps develop software efficiently and with high quality.
Process definition means specifying:
•**What** will be done
•**How** it will be done
•**Who** will do it
•**When** it will be done
To manage software development effectively, process definitions are organized into **levels**:
1.Organizational Level
2.Project Level
3.Activity / Task Level

Levels of Process Definition

Why Define Levels of Process?

- Helps manage complex software development
- Clearly defines roles and responsibilities
- Ensures standardization and quality
- Helps in monitoring progress and controlling changes

In short, levels of process definition provide clarity, consistency, and repeatability.

Organizational Level (High Level)

•Defines the overall process framework for the entire organization
•Ensures all projects follow standard processes
•Provides a high-level view of development activities

Characteristics:

•High-level, broad view of software processes
•Organization-wide standards and policies
•Generic, not project-specific
•Defines quality assurance rules, coding standards, and tool usage

Organizational Level (High Level)

Example:
A software company defines its standard SDLC process for all projects:
1.Requirements Gathering
2.System Design
3.Implementation / Coding
4.Testing
5.Deployment
Tools and Standards:
•Version Control (Git)
•Bug Tracking (JIRA)
•Coding Standards (PEP-8 for Python)

Project Level (Medium Level)

Definition:

This level tailors the **organization's high-level process** for a **specific project**.

Characteristics:

- Medium-level process
- Defines project-specific timelines, tools, and team roles
- Ensures project objectives meet organizational standards

Example:

University Management System Project:

- Requirements: 2 weeks
- Design: 1 week
- Coding: 4 weeks
- Testing: 2 weeks
- Deployment: 1 week

Tools: MySQL, VS Code, JIRA

Activity / Task Level (Low Level)

Definition:

This is the **most detailed level**. It specifies **exact steps for individual activities** in software development.

Characteristics:

•Low-level, highly detailed

•Defines exact tasks, roles, documentation, and approvals

•Ensures repeatability and quality

Example:

Coding Activity:

1.Write code for login page

2.Peer code review

3.Unit testing

4.Debugging

5.Submit for integration

Testing Activity:

1.Prepare test cases

2.Execute tests

3.Log defects

4.Retest after fixes

What is Software Process Implementation

- The process of putting a software process into action in a real project
- Ensures software development activities are performed as planned
- Bridges the gap between process design and execution
- Focuses on practical deployment of standards, methods, and tools

Implementation Steps Overview

- **Planning** – Scope, resources, schedule
- **Process Deployment** – Put process into practice
- **Training & Awareness** – Educate team members
- **Monitoring & Measurement** – Track compliance and performance
- **Feedback & Adjustment** – Improve process iteratively
- **Documentation** – Record procedures, reports, lessons learned

Step 1 – Planning

- Identify **objectives and goals** of implementation
- Select **process model/framework** (e.g., Waterfall, Agile, Incremental)
- Allocate **resources and responsibilities**
- Define **timeline, milestones, and risk management strategies**

Example: A team decides to implement Agile for a mobile app project with sprints of 2 weeks each.

Step 2 – Process Deployment

- **Apply the defined process** in real project activities
- Assign **roles and responsibilities** according to the process
- Use **standard tools, templates, and methods**
- Ensure **all team members follow the same workflow**

Example:

- Developers follow Agile board and daily standups
- QA follows test case templates and defect tracking

Step 3 – Training & Awareness

•Conduct **training sessions for staff** on process steps and tools

•Ensure **awareness of standards, quality measures, and reporting**

•Provide **guidelines, manuals, and reference documents**

Example:

•Team workshop on Git usage, Agile ceremonies, and coding standards

Step 4 – Monitoring & Measurement

- Track **process compliance**
- Measure **key performance indicators (KPIs)** like defect density, task completion time
- Use **tools for reporting**: dashboards, reports, metrics

Example:

- Track sprint velocity in Agile
- Track bug closure rates in QA

Step 5 – Feedback & Adjustment

- Gather **feedback from team and stakeholders**
- Identify **process bottlenecks and inefficiencies**
- Adjust the process for **better performance in future projects**
- Apply **lessons learned** in next iteration

Example:

- Team realizes testing phase is delayed → add automated testing in next sprint

Step 6 – Documentation

- Maintain **process documents, manuals, and templates**
- Record **lessons learned and process improvements**
- Provide a **reference for future projects**

Example:

- Document Agile workflow, coding standards, sprint retrospectives

Implementation Methods

Common Methods to Implement Software Processes

- **Top-down (Directive):** Management enforces process across teams
- **Bottom-up (Participatory):** Team members gradually adopt and suggest improvements
- **Pilot Project:** Start process on one project, then scale
- **Phased Rollout:** Introduce process in **stages or modules**

Challenges in Process Implementation

- **Resistance to change** from team members
- Lack of **skills or training**
- Poor **communication or awareness**
- Tools not integrated properly
- **Time constraints** or tight deadlines